

BENIGN PROSTATIC HYPERPLASIA

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APPROACH TO LOWER URINARY TRACT SYMPTOMS (LUTS)

HISTORY

- **Nature:**

- Voiding Symptoms

- Hesitancy

- Straining

- Double voiding

- Sensation of incomplete emptying

- Weak flow

- Terminal dribbling

- Storage Symptoms

- Frequency

- Urgency

- Nocturia

- **Duration of Symptoms**

- **Other Associated Symptoms:**

- Gross haematuria, urinary incontinence - any need for diapers?, dysuria, symptoms of UTI

- **Medication List:**

- Diuretics, SGLT2-inhibitors, anti-psychotics, traditional medications, supplements

- **Past Surgical History:**

- Procedures like indwelling catheter, cystoscopy, bladder stones, TURP

- **Past Medical History:**

- DM, CCF, Parkinson's, spinal injuries

- **Social History:**

- Coffee, tea, alcohol drinking, smoking

Red Flags from History: Haematuria, acute retention of urine, fever with loin pain

Differentials for LUTS in males:

- UTI
- Prostatitis
- Distal ureteral stone
- Ureteral stricture
- Bladder tumour
- Bladder issues: Neurogenic bladder dysfunction, Over Active Bladder (OAB) - Detrusor over-activity, detrusor under-activity
- Nocturnal polyuria
- Foreign body

PHYSICAL EXAMINATION

BP and hydration status

Abdominal: Mass, palpable or percussable bladder, inguinal hernia, ballotable kidney
Any signs of ESRF.

Digital Rectal Examination (DRE)

Prostate: Size, consistency, median sulcus, tenderness

****Red flags:** Irregular, hard, nodules

Rectal mass

Assess anal tone - Poor anal tone and sacral anaesthesia suggest possible neurogenic voiding dysfunction.

Penis: Phimosis, hypospadias

INVESTIGATION

- UFEME to screen for pyuria, haematuria, proteinuria or other pathology
Urine culture if UFEME abnormal.
- Baseline PSA
Levels of PSA was proposed to be a good surrogate for estimating prostate volume. For prostate volume > 30 g, PSA would be > 1.5 µg/L. For patients who presents with LUTS, PSA testing helps to detect prostate cancer and prostatitis, especially when DRE reveals an abnormal prostate.

PSA Screening (Singapore Urological Association, 2017)

Population screening for prostate cancer with PSA is controversial and not recommended.

For men between 50-70 years of age with a life expectancy of more than 10 years, PSA screening may be offered after an informed discussion with patient on potential benefits and harms including the possibilities of false positive and false negative results, complications of subsequent TRUS guided biopsy and false negative biopsies.

Men with a strong family history of prostate cancer, i.e., one or more first-degree relatives diagnosed before the age of 65 years, may be offered screening 5 to 10 years younger than the youngest prostate cancer in the family.

Routine screening in men younger than 50 year old without a strong family history of prostate cancer is not recommended.

Men who have life expectancy of less than 10-15 years should be informed that testing and treatment is unlikely to be beneficial.

A PSA value of $> 4 \mu\text{g/L}$ would need referral to urology for further evaluation.

To reduce the harms of screening, a routine screening interval of two years or more may be preferred over annual screening in asymptomatic men who have participated in shared decision-making and decided on screening. As compared to annual screening, it is expected that screening intervals of two years preserve the majority of the benefits and reduce over diagnosis and false positives. For patients who were previously on follow up with urology, the interval of PSA screening should be deferred to the advice of the urologist returning the patient to primary care follow up.

Voiding Diary

Should be used when frequency, urgency or nocturia is the dominant symptoms.

Helps to identify patients with isolated nocturnal polyuria, excessive fluid intake or overactive bladder.

INTERNATIONAL PROSTATE SYMPTOMS SCORE (IPSS) AND QUALITY OF LIFE SCORE (QOL)

Patient Name: Date done:	Not at all	Less than 1 time in 5	Less than half the time	About half the time	More than half the time	Almost always	Score
Incomplete Emptying Over the past month, how often have you had a sensation of not emptying your bladder completely after you finished urinating?	0	1	2	3	4	5	
Frequency Over the past month, how often have you had to urinate again less than 2 hours after you finished urinating?	0	1	2	3	4	5	
Intermittency Over the past month, how often have you found you stopped and started again several times when you urinated?	0	1	2	3	4	5	
Urgency Over the past month, how often have you found it difficult to postpone urination?	0	1	2	3	4	5	
Weak Stream Over the past month, how often you had a weak urinary stream?	0	1	2	3	4	5	
Straining Over the past month, how often have you had to push or strain to begin urination?	0	1	2	3	4	5	
	None	1 time	2 times	3 times	4 times	5 or more times	
Nocturia Over the past month, how many times did you most typically get up to urinate from the time you went to bed at night until the time you got up in the morning?	0	1	2	3	4	5	
Your total IPSS score							
Quality of Life Due to Urinary Symptoms	Delighted	Pleased	Mostly satisfied	Mixed mostly	Mostly dissatisfied	Unhappy	Terrible
If you were to spend the rest of your life with your urinary condition just the way it is now, how would you feel about that?	0	1	2	3	4	5	6

Scoring interpretation:

IPSS

0 - 7: Mild

8 - 19: Moderate

20 - 35: Severe

QOL

< 3 not bothered

≥ 3 bothered

Further Investigation

These investigations may be required in patients with a definite indication, such as gross haematuria, uncertain diagnosis, DRE abnormalities, poor response to medical therapy or for surgical planning.

- Urine cytology
- Uroflowmetry
- Urodynamics
- Cystoscopy
- Radiological evaluation of upper urinary tract, e.g., IVU, CT KUB, Urogram
- Prostate ultrasound
- MRI prostate and prostate biopsy are indicated when the PSA is elevated

MANAGEMENT

Medical Therapy:

- Alpha-1 blockers (Acts by relaxation of smooth muscle in prostatic urethra, bladder neck and blood vessels, IPSS decreases by about 35 -40% and the flow rate by 20-25%.) However, they do not prevent progression of BPH. Patients usually experience full therapeutic effect of alpha-1 blockers within one week. **We recommend initiating Alpha-1 blockers in patients whose urinary symptoms affect and limit their function.**

	Dose	Subsidy	Cost	Risk of postural hypotension	Risk of retrograde ejaculation	Risk of floppy iris syndrome	Remarks
Alfuzoxin XL (Xatral XL)	10 mg ON	SDL2	\$	Medium	Low	Medium	Recommended for younger patients
Tamsulosin (Harnal)	0.4 mg ON	NS	\$\$	Low	Medium	High	Recommended in elderly with high falls risk
Terazosin (Hytrin)	1 - 10mg ON	SDL2	2mg: \$ 5mg: \$\$\$	High	Low	Low	Suitable as an anti-hypertensive. Can be increased up to 20mg if inadequate response after 4-6 weeks

^ Cost \$ <\$10 per month \$\$ \$10 - <\$20 per month \$\$\$ \$20 - <\$30 per month
 \$\$\$\$ \$30 - <\$60 per month \$\$\$\$\$ \$60 - <\$90 per month \$\$\$\$\$\$ \$90 or more per month

Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Non-Standard Drug NS)

Unit prices before GST and accurate as of Jan 2024

Prices shown are an estimate. Please contact pharmacy for accurate pricing.

- 5-Alpha-reductase inhibitors (5ARI) are indicated for prostate volumes >30 g and significant obstruction. Efficacy is more pronounced in those with larger prostatic volumes. Efficacy is minimal for prostatic volume < 30 g. (Decreases prostate volume by about 18 - 28% after 6 - 12 months, IPSS decreases by about 20 - 30%, reduces PSA by 50% after 6 - 12 months: suggested to have a baseline PSA before treatment). Patients usually experience full therapeutic effects after six months of treatment.
 - Finasteride (Proscar): 5 mg OD
 - Dutasteride (Avodart): 0.5 mg OD

ADR

Decreased libido, erectile dysfunction, ejaculatory disorders, mastalgia, gynaecomastia.

General guidance is that the PSA value should reduce by 50% after at least 6 months of therapy of 5ARI and generally be maintained at that level as long as the patient continues to take the medication regularly. Hence a red flag will be that a patient's PSA reduces approximately by half, but then begins to rise for an unexplained reason, assuming he was compliant with the medication.

Other Managements: Examples of Lifestyle Modifications

- Modify drinking habits, cut down fluid intake at night
- Restriction of caffeine and alcohol intake
- Avoidance / monitoring usage of some drugs (diuretics, antihistamines, antidepressants)
- Stop smoking
- Regular exercise
- Lose weight if BMI is high
- Timed or organised voiding (Bladder retraining)
- Keep other medical conditions well managed

Surgical Therapy Options

- Bipolar Transurethral Resection of Prostate (TURP)
- Enucleation of obstructing prostatic adenoma
- Transurethral Laser prostatectomy
- Transurethral Incision of Prostate (TUIP) Open prostatectomy
- UroLift and Rezum water vapour treatments

REFERRAL TO UROLOGIST

- Persistent LUTS (based on moderate - severe IPSS or QOL score ≥ 3)
- Gross haematuria
- Urinary incontinence
- Palpable bladder
- DRE suspicious of prostate cancer
- Abnormal PSA levels (> 4.0 ng/ml)
- A rise in PSA while on 5-alpha reductase inhibitors
- Recurrent infection
- Complications arising from obstruction like hydronephrosis, renal failure
- History / risk of urethral stricture
- Neurological disease raising the likelihood of a primary bladder disorder
- Failure of medical treatment at primary care level

Note: Post catheterised patients presenting with ARU (acute retention of urine) - To consider A&E referral (if gross haematuria or signs of urosepsis) or Direct Access referral to Urology (within 1 week, if patient is assessed to be able to take care of urinary catheter / bag)

FOLLOW UP VISIT IN PRIMARY CARE

- To assess treatment success or failure and possible adverse events.
- The time after initiation of therapy for the assessment of treatment success varies according to the pharmacological treatment prescribed and is usually 2 to 4 weeks for alpha blocker therapy and at least 3 months for a 5 α -reductase inhibitor.
- If treatment is successful and the patient is satisfied, annual PSA for patients below the age of 70 years is generally recommended for those on 5-alpha reductase inhibitors.
- The follow-up strategy will allow the physician to detect any changes that have occurred in the last year, more specifically, if symptoms have progressed or become more bothersome, or if a complication has developed creating an indication imperative for surgery.

RECOMMENDED CARE COMPONENTS FOR BENIGN PROSTATIC HYPERPLASIA

Recommended Care Components	Minimum Frequency*	Remarks
Review of Lower Urinary Tract Symptoms	Annually	Recommended tool for assessment of LUTS is the - International Prostate Symptom / Quality of Life Score
Clinical Examination - Abdominal and Digital Rectal Exam	Initial assessment	Abdominal examination includes assessment for a palpable bladder. Rectal examination to assess size, consistency and regularity of prostate
Co-Morbidity Assessment (includes medication review)	Initial assessment	
Urine Dipstick or Microscopy [#]	Initial assessment	Screen for haematuria, pyuria and glycosuria

** More frequently if clinically indicated*

This is a non-reportable clinical indicator

REFERENCES

1. Singapore Urological Association Clinical Guidelines for Male Lower Urinary Tract Symptoms/Benign Prostatic Hyperplasia. Singapore Med J. 2017 Aug; 58(8): 473-480
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